

Lab & Activities Week 06 Agent Fundamentals

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Activity 1 · Warm-up Concept Quiz

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10 MININDIVIDUALFORMATIVE

1. In the Perception-Reasoning-Action loop, what turns an "Action" (a tool result) back into new input for the model?

- ☐ The model re-reads its own previous answer
- ☒ Perception — the tool's result is fed back in as new context
- ☐ Nothing; each loop iteration is independent

Correct answer: B. The observation becomes the next perception step, closing the loop.

2. Who actually executes the Python function when an LLM "calls a tool"?

- ☐ The LLM provider's servers
- ☐ The function executes itself automatically once described
- ☒ Your application code, after receiving the model's structured request

Correct answer: C. The model only ever emits a request; your code is responsible for running it and returning the result.

3. Your agent's tool has the docstring "handles topic stuff". It keeps getting ignored or misused. Why?

- ☒ The description is the model's only signal for when/how to use the tool, and it's too vague
- ☐ The function name is too short
- ☐ Tools need to return strings, not dicts

Correct answer: A. Treat tool descriptions as the API documentation the model relies on.

4. A task has a clear, predictable sequence of steps known in advance. Which pattern minimizes LLM reasoning calls?

- ☐ ReAct
- ☒ Plan-and-Execute
- ☐ Neither; always use raw prompting

Correct answer: B. Plan-and-Execute pays for one planning call, then runs steps more cheaply.

Check answers

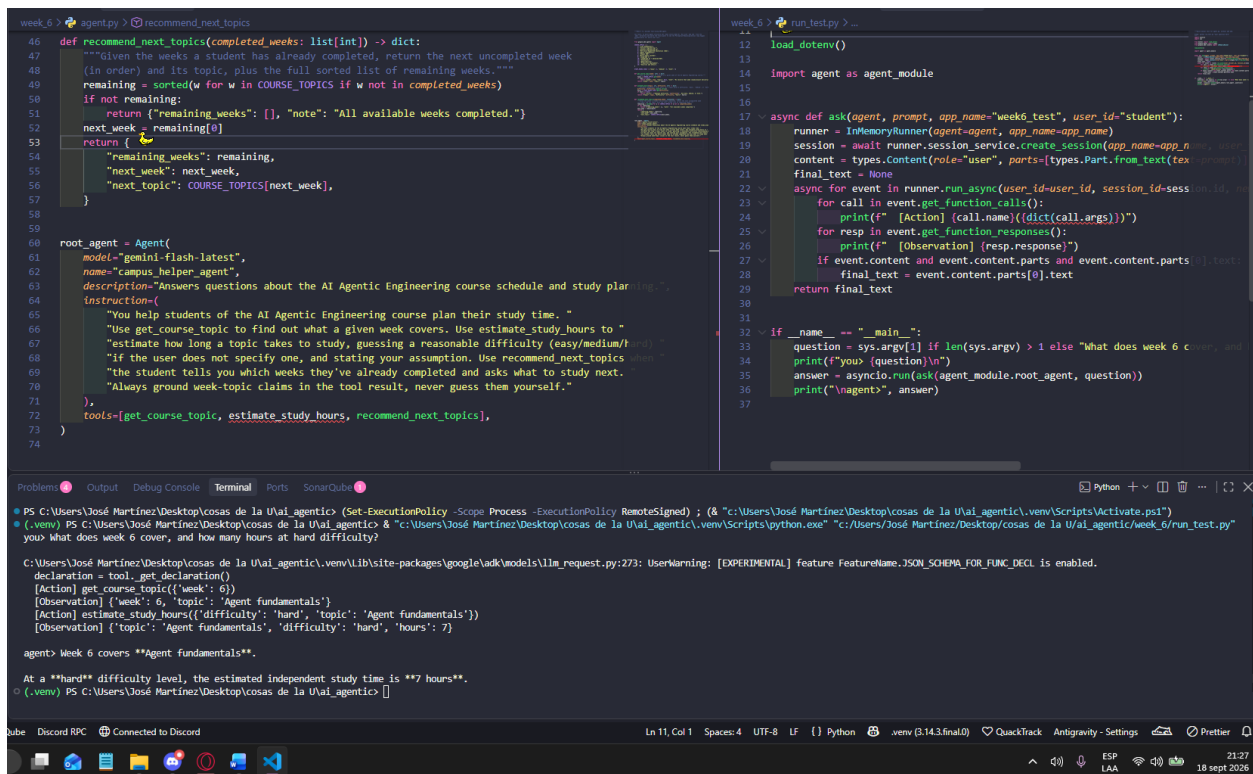
Score: 4 / 4

Activity 2 · Instructor-Led Live Coding

Followed the manual ReAct loop idea first (no separate content page file was in the repo, so I reconstructed it directly from the two tools in agent.py/app.py), then intentionally tried to break a tool exactly as suggested, before running both runnable apps.

Breaking a tool on purpose

First attempt: renamed the estimate_study_hours function to estimate_hours in the tools list, but left the OLD name ("estimate_study_hours") in the system instruction text -- the classic "renamed it, forgot a call site" bug the activity describes.



The screenshot shows a VS Code editor with two Python files open. The left file, `agent.py`, defines a `recommend_next_topics` function and a `root_agent` class. The right file, `run_test.py`, contains a `main` function that interacts with the agent. The terminal at the bottom shows the output of the `run_test.py` script, including a warning about an experimental feature and the agent's response to a query about week 6.

```
week_6 > agent.py > recommend_next_topics
46 def recommend_next_topics(completed_weeks: list[int]) -> dict:
47     """Given the weeks a student has already completed, return the next uncompleted week
48     (in order) and its topic, plus the full sorted list of remaining weeks."""
49     remaining = sorted(w for w in COURSE_TOPICS if w not in completed_weeks)
50     if not remaining:
51         return {"remaining_weeks": [], "note": "All available weeks completed."}
52     next_week = remaining[0]
53     return {
54         "remaining_weeks": remaining,
55         "next_week": next_week,
56         "next_topic": COURSE_TOPICS[next_week],
57     }
58
59
60 root_agent = Agent(
61     model="gemini-flash-latest",
62     name="campus_helper_agent",
63     description="Answers questions about the AI Agentic Engineering course schedule and study planning.",
64     instruction=(
65         "You help students of the AI Agentic Engineering course plan their study time. "
66         "Use get_course_topic to find out what a given week covers. Use estimate_study_hours to "
67         "estimate how long a topic takes to study, guessing a reasonable difficulty (easy/medium/hard). "
68         "If the user does not specify one, and stating your assumption. Use recommend_next_topics when "
69         "the student tells you which weeks they've already completed and asks what to study next. "
70         "Always ground week-topic claims in the tool result, never guess them yourself."
71     ),
72     tools=[get_course_topic, estimate_study_hours, recommend_next_topics],
73 )
74
```

```
week_6 > run_test.py > ...
12 load_dotenv()
13
14 import agent as agent_module
15
16
17
18 async def ask(agent, prompt, app_name="week6 test", user_id="student"):
19     runner = InMemoryRunner(agent=agent, app_name=app_name)
20     session = await runner.session_service.create_session(app_name=app_name, user_id=user_id)
21     content = types.Content(role="user", parts=[types.Part.from_text(text=prompt)])
22     final_text = None
23     async for event in runner.run_async(user_id=user_id, session_id=session_id, content=content):
24         for call in event.get_function_calls():
25             print(f" [Action] (call.name):{dict(call.args)}")
26             for resp in event.get_function_responses():
27                 print(f" [Observation] (resp.response):")
28                 if event.content and event.content.parts and event.content.parts[0].text:
29                     final_text = event.content.parts[0].text
30     return final_text
31
32 if __name__ == "__main__":
33     question = sys.argv[1] if len(sys.argv) > 1 else "What does week 6 cover, and how many hours does it take to study?"
34     print(f">you> {question}\n")
35     answer = asyncio.run(ask(agent_module.root_agent, question))
36     print(f">agent>, {answer}")
37
```

Problems Output Debug Console Terminal Ports SonarQube

```
PS C:\Users\José Martínez\Desktop\cosas de la Uai_agentic> (Set-ExecutionPolicy -Scope Process -ExecutionPolicy RemoteSigned) ; (& "c:\Users\José Martínez\Desktop\cosas de la Uai_agentic\venv\Scripts\Activate.ps1")
(.venv) PS C:\Users\José Martínez\Desktop\cosas de la Uai_agentic> & "c:\Users\José Martínez\Desktop\cosas de la Uai_agentic\venv\Scripts\python.exe" "c:\Users\José Martínez\Desktop\cosas de la Uai_agentic\week_6\run_test.py"
you> What does week 6 cover, and how many hours at hard difficulty?

C:\Users\José Martínez\Desktop\cosas de la Uai_agentic\venv\Lib\site-packages\google\adk\models\llm_request.py:273: UserWarning: [EXPERIMENTAL] feature FeatureName.JSON_SCHEMA_FOR_FUNC_DECL is enabled.
  declaration = tool.get_declaration()
[Action] get_course_topic({'week': 6})
[Observation] {'week': 6, 'topic': 'Agent fundamentals'}
[Action] estimate_study_hours({'difficulty': 'hard', 'topic': 'Agent fundamentals'})
[Observation] {'topic': 'Agent fundamentals', 'difficulty': 'hard', 'hours': 7}

agent> Week 6 covers **Agent fundamentals**.

At a **hard** difficulty level, the estimated independent study time is **7 hours**.
(.venv) PS C:\Users\José Martínez\Desktop\cosas de la Uai_agentic>
```

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This did NOT break anything -- the model correctly called estimate_hours (the real, current tool), completely ignoring the stale name in the instruction text. ADK only ever exposes the tool's real, introspected name/schema to the model, so a rename that's inconsistent only in prose is harmless dead text, never a runtime bug.

Second attempt, a genuine break: left the tool's registered name alone, but broke its INTERNAL body instead -- one line inside `estimate_study_hours` references `STUDY_HOURS_TABLE_RENAMED`, a name that doesn't exist (the real table is still called `STUDY_HOURS_TABLE`). This is what "renamed it, forgot a call site" actually looks like as a real Python bug.

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week 6 > agent.py ...
46 def recommend_next_topics(completed_weeks: list[int]) -> dict:
47     remaining_topics = [t for t in COURSE_TOPICS if t not in completed_weeks]
48     if not remaining:
49         return {"remaining_weeks": [], "note": "All available weeks completed."}
50     next_week = remaining[0]
51     return {
52         "remaining_weeks": remaining,
53         "next_week": next_week,
54         "next_topic": COURSE_TOPICS[next_week],
55     }
56
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58
59 root_agent = Agent(
60     model="gemini-flash-latest",
61     name="campus helper agent",
62     description="Answers questions about the AI Agentic Engineering course schedule and study planning.",
63     instruction=
64         "You help students of the AI Agentic Engineering course plan their study time. "
65         "Use get_course_topic to find out what a given week covers. Use estimate_study_hours to "
66         "estimate how long a topic takes to study, guessing a reasonable difficulty (easy/medium/hard). "
67         "If the user does not specify one, and stating your assumption. Use recommend_next_topics when "
68         "the student tells you which weeks they've already completed and asks what to study next. "
69         "Always ground week-topic claims in the tool result, never guess them yourself.",
70 ),
71 tools=[get_course_topic, estimate_hours, recommend_next_topics],
72 )
73
Variable de entorno en w... x run_test.py 2 x ...
week 6 > run_test.py > ask
11 load_dotenv()
12
13 import agent as agent_module
14
15
16
17 async def ask(agent, prompt, app_name="week6_test", user_id="student"):
18     runner = InMemoryRunner(agent=agent, app_name=app_name)
19     session = await runner.session_service.create_session(app_name=app_name, user_id=user_id)
20     content = types.content(role="user", parts=[types.Part.from_text(text=prompt)])
21     final_text = None
22     async for event in runner.run_async(user_id=user_id, session_id=session_id, content=content):
23         for call in event.get_function_calls():
24             print(f" [Action] {call.name}({dict(call.args)})")
25         for resp in event.get_function_responses():
26             print(f" [Observation] {resp.response}")
27         if event.content and event.content.parts and event.content.parts[0].text:
28             final_text = event.content.parts[0].text
29     return final_text
30
31
32 if __name__ == "__main__":
33     question = sys.argv[1] if len(sys.argv) > 1 else "What does week 6 cover?"
34     print(f"you: {question}\n")
35     answer = asyncio.run(agent_module.root_agent, question))
36     print(f"\nagent>: {answer}")
37
Problems Output Debug Console Terminal Ports SonarQube
Python + + + + +
(.venv) PS C:\Users\José Martínez\Desktop\cosas de la Uai\agentic> "c:\Users\José Martínez\Desktop\cosas de la Uai\agentic\venv\Scripts\python.exe" "c:\Users\José Martínez\Desktop\cosas de la Uai\agentic\week 6\run_test.py"
NameError: name 'estimate_study_hours' is not defined. Did you mean: 'estimate_hours'?
(.venv) PS C:\Users\José Martínez\Desktop\cosas de la Uai\agentic> "c:\Users\José Martínez\Desktop\cosas de la Uai\agentic\venv\Scripts\python.exe" "c:\Users\José Martínez\Desktop\cosas de la Uai\agentic\week 6\run_test.py"
you: What does week 6 cover, and how many hours at hard difficulty?
C:\Users\José Martínez\Desktop\cosas de la Uai\agentic\venv\Lib\site-packages\google\adk\models\llm_request.py:273: UserWarning: [EXPERIMENTAL] feature FeatureName.JSON_SCHEMA_FOR_FUNC_DECL is enabled.
  declaration = tool.get_declaration()
[Action] get_course_topic({'week': 6})
[Observation] {'week': 6, 'topic': 'Agent fundamentals'}
[Action] estimate_hours({'difficulty': 'hard', 'topic': 'Agent fundamentals'})
Node execution failed with exception
Traceback (most recent call last):
  File "C:\Users\José Martínez\Desktop\cosas de la Uai\agentic\venv\Lib\site-packages\google\adk\workflow_node_runner.py", line 136, in run
    await self._execute_node(ctx, node_input)
  File "C:\Users\José Martínez\Desktop\cosas de la Uai\agentic\venv\Lib\site-packages\google\adk\workflow_node_runner.py", line 274, in _execute_node
    await self._run_node_loop(ctx, node_input)
  File "C:\Users\José Martínez\Desktop\cosas de la Uai\agentic\venv\Lib\site-packages\google\adk\workflow_node_runner.py", line 288, in _run_node_loop
    async for event in agen:
  File "C:\Users\José Martínez\Desktop\cosas de la Uai\agentic\venv\Lib\site-packages\google\adk\workflow_base_node.py", line 166, in run
    self._track_event_in_context(event, ctx)
  File "C:\Users\José Martínez\Desktop\cosas de la Uai\agentic\venv\Lib\site-packages\google\adk\workflow_base_node.py", line 166, in run
```

```
week 6 > agent.py > recommend next topics
46 def recommend_next_topics(completed_weeks: list[int]) -> dict:
47     remaining = sum(1 for w in COURSE_TOPICS if w not in completed_weeks)
48     if not remaining:
49         return {"remaining_weeks": [], "note": "All available weeks completed."}
50     next_week = remaining[0]
51     return {
52         "remaining_weeks": remaining,
53         "next_week": next_week,
54         "next_topic": COURSE_TOPICS[next_week],
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60 root_agent = Agent(
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65         "You help students of the AI Agentic Engineering course plan their study time. "
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week 6 > run_test.py > ask
12 load_dotenv()
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14 import agent as agent_module
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17 async def ask(agent, prompt, app_name="week6_test", user_id="student"):
18     runner = InMemoryRunner(agent=agent, app_name=app_name)
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20     content = types.Content(role="user", parts=[types.Part.from_text(text=prompt)])
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22     async for event in runner.run_async(user_id=session_id, session_id=session_id):
23         for call in event.get_function_calls():
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27                 if event.content and event.content.parts and event.content.parts[0].text:
28                     final_text = event.content.parts[0].text
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32 if __name__ == "__main__":
33     question = sys.argv[1] if len(sys.argv) > 1 else "What does week 6 cover, and how many hours at hard difficulty?"
34     print(f"you> {question}\n")
35     answer = asyncio.run(ask(agent_module.root_agent, question))
36     print("\nagents", answer)
37

Problem Output Debug Console Terminal Ports SonarQube
(.venv) PS C:\Users\José Martínez\Desktop\cosas de la Uai_agentic> "c:\Users\José Martínez\Desktop\cosas de la Uai_agentic\.venv\Scripts\python.exe" "c:\Users\José Martínez\Desktop\cosas de la Uai_agentic\week_6/run_test.py"
you> What does week 6 cover, and how many hours at hard difficulty?
C:\Users\José Martínez\Desktop\cosas de la Uai_agentic\.venv\Lib\site-packages\google\ads\models\llm_request.py:273: UserWarning: [EXPERIMENTAL] feature FeatureName.JSON_SCHEMA_FOR_FUNC_DECL is enabled.
  declaration = tool.get_declaration()
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[Action] estimate_study_hours({'difficulty': 'hard', 'topic': 'Agent fundamentals'})
[Observation] {'topic': 'Agent fundamentals', 'difficulty': 'hard', 'hours': 7}
agents> week 6 covers **Agent fundamentals**.
At a **hard** difficulty level, it is estimated to take **7 hours** of independent study.
(.venv) PS C:\Users\José Martínez\Desktop\cosas de la Uai_agentic>
```

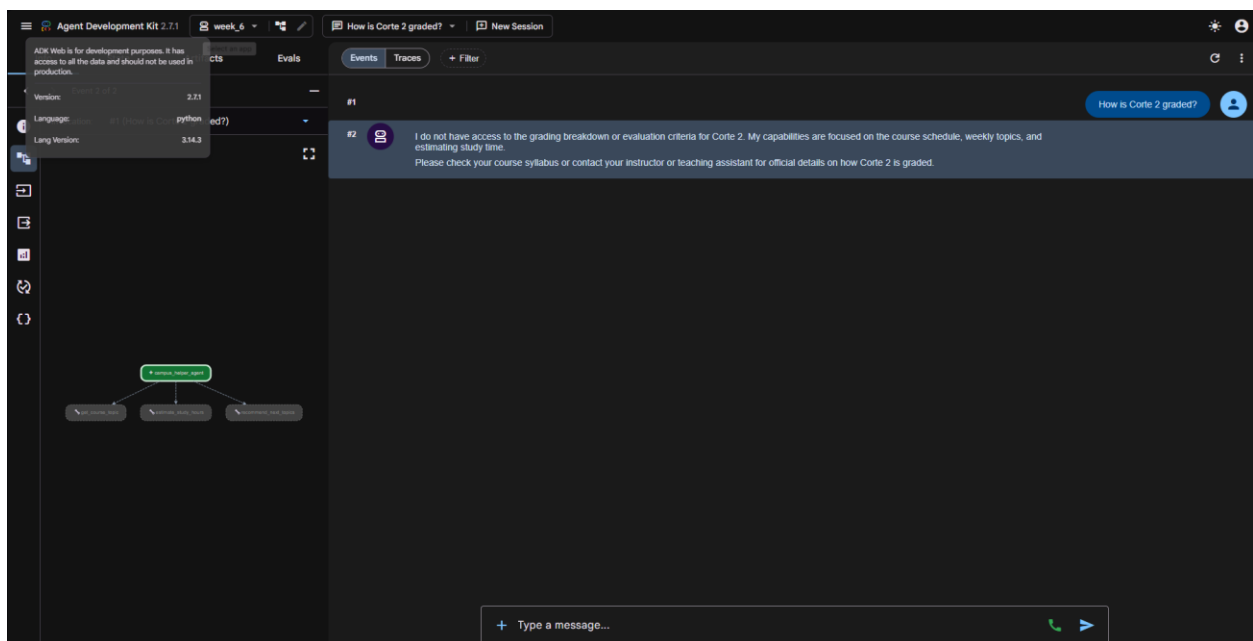
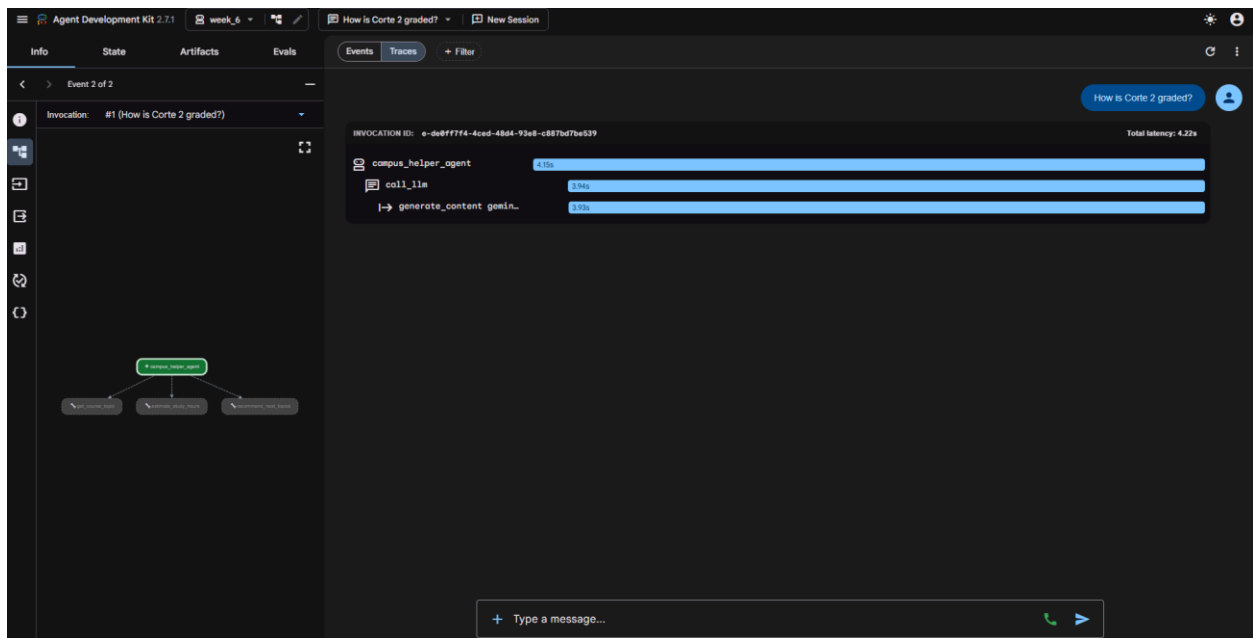
Neither framework turns an unhandled exception inside a tool into a graceful "Observation: error, please retry" -- both just crash the entire run with the raw traceback. The self-correction the activity describes only happens if the tool itself catches the exception and returns a structured error dict (e.g. {"error": "..."} -- that translation is the tool author's job, not something either framework does automatically. Full detail in week_6/NOTES.md.

Running the two runnable apps

adk/campus_helper_agent (via InMemoryRunner, same pattern used in Weeks 2-4) and langgraph/app.py (create_react_agent), both asked "How is Corte 2 graded?"-style logistics questions, worked as expected end to end -- full traces are in Activity 4, where the same two apps are exercised with the new third tool added.

```
week_6 agent.py > recommend_next_topics
46 def recommend_next_topics(completed_weeks: list[int]) -> dict:
47     remaining = [topic for topic in COURSE_TOPICS if topic not in completed_weeks]
48     if not remaining:
49         return {"remaining_weeks": [], "note": "All available weeks completed."}
50     next_week = remaining[0]
51     return {
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53         "next_week": next_week,
54         "next_topic": COURSE_TOPICS[next_week],
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60 root_agent = Agent(
61     model="gemini-flash-latest",
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The screenshot displays the Agent Development Kit (ADK) 2.7.1 web interface. The top navigation bar includes a browser address bar with the URL '127.0.0.1:8000/dev-ui/' and a 'NEW SESSION' button. Below this, a sidebar on the left contains tabs for 'Info', 'State', 'Artifacts', and 'Evals'. The main content area is divided into 'Events' and 'Traces' sections, with a '+ Filter' button. A large, empty dark gray area occupies the center of the screen, with a faint instruction 'Select an event or trace span to view details' at the top left. At the bottom, there is a message input field with a plus icon and the text '+ Type a message...'. The bottom status bar shows system icons and the time '21:45 18 sept 2026'.



On terminal

Thought: Difficulty wasn't given. I'll assume "medium" for all three and say so.

Action: `estimate_study_hours(topic="Agent fundamentals", difficulty="medium")`

Observation: `{"topic": "Agent fundamentals", "difficulty": "medium", "hours": 4}`

Action: `estimate_study_hours(topic="Google ADK", difficulty="medium")`

Observation: `{"topic": "Google ADK", "difficulty": "medium", "hours": 4}`

Action: `estimate_study_hours(topic="Multi-agent systems", difficulty="medium")`

Observation: `{"topic": "Multi-agent systems", "difficulty": "medium", "hours": 4}`

Final answer: Assuming medium difficulty for all three (none was specified), budget about 12 hours total (4 each) for weeks 6-8. The tools can't actually tell you which one is "hardest" -- `estimate_study_hours` only returns a number for the difficulty you hand it, so guessing the same difficulty for all three makes them tie by construction. Answering "which is hardest" honestly would require asking the user to rate each topic's difficulty individually, not guessing one value for all three.

2. A shortcut without losing correctness

The three `get_course_topic` calls are fully independent of each other (none needs another's result), so they don't have to be issued one at a time -- they can be batched into the same reasoning step. This isn't hypothetical: it's exactly what was observed empirically in Activity 4, where both ADK and LangGraph issued two independent tool calls together in a single step rather than strictly serially.

3. Where ReAct's step-by-step nature helps

If one of the three weeks were invalid (say, week 5 or week 11 -- no lecture that week), `get_course_topic` would come back with `topic: null` and a note instead of a real topic. A rigid upfront plan would have already committed to calling `estimate_study_hours(topic=None, ...)` before ever seeing

that result. ReAct's loop lets the agent see each Observation before deciding the next Action, so it can skip estimating hours for a week that doesn't exist instead of blindly executing a fixed 6-step plan.

Activity 4 · Individual Lab: Add a Tool, Both Ways

Added `recommend_next_topics(completed_weeks: list[int]) -> dict` to both `agent.py` (ADK) and `app.py` (LangGraph) -- given the weeks a student already finished, it returns the next uncompleted week and its topic. Chosen because it chains naturally with `estimate_study_hours` (and, in one test, with `get_course_topic` too), which the activity requires.

```
def recommend_next_topics(completed_weeks: list[int]) -> dict:
    """Given the weeks a student has already completed, return the next
    uncompleted week (in order) and its topic, plus the full sorted list
    of remaining weeks."""
    remaining = sorted(w for w in COURSE_TOPICS if w not in completed_weeks)
    if not remaining:
        return {"remaining_weeks": [], "note": "All available weeks completed."}
    next_week = remaining[0]
    return {
        "remaining_weeks": remaining,
        "next_week": next_week,
        "next_topic": COURSE_TOPICS[next_week],
    }
```

3 test prompts, run for real on both frameworks

Prompt 1: I've completed weeks 1 through 4. What should I study next, and how many hours should I budget if it's a hard topic?

ADK:

```
[Action] recommend_next_topics({'completed_weeks': [1, 2, 3, 4]})

[Observation] {'remaining_weeks': [6, 7, 8, 9, 10, 12, 13, 14, 15], 'next_week': 6, 'next_topic': 'Agent
fundamentals'}

[Action] estimate_study_hours({'topic': 'Agent fundamentals', 'difficulty': 'hard'})

[Observation] {'topic': 'Agent fundamentals', 'difficulty': 'hard', 'hours': 7}

[Final] Next up is Week 6 (Agent fundamentals). At hard difficulty, budget ~7 hours.
```

LangGraph:

```
[Action] recommend_next_topics({'completed_weeks': [1, 2, 3, 4]})

[Observation] {"remaining_weeks": [6,7,8,9,10,12,13,14,15], "next_week": 6, "next_topic": "Agent
fundamentals"}

[Action] estimate_study_hours({'difficulty': 'hard', 'topic': 'Agent fundamentals'})

[Observation] {"topic": "Agent fundamentals", "difficulty": "hard", "hours": 7}

[Final] Same content, wrapped in LangChain's block-list format (see NOTES.md).
```

Prompt 2: What does week 9 cover, and how many hours should I budget for it at medium difficulty?

ADK:

```
[Action] get_course_topic({'week': 9})
```

```
[Observation] {'week': 9, 'topic': 'LangGraph I'}

[Action] estimate_study_hours({'topic': 'LangGraph I', 'difficulty': 'medium'})

[Observation] {'topic': 'LangGraph I', 'difficulty': 'medium', 'hours': 4}

[Final] Week 9 covers LangGraph I. At medium difficulty, budget ~4 hours.
```

LangGraph:

```
[Action] get_course_topic({'week': 9})

[Observation] {"week": 9, "topic": "LangGraph I"}

[Action] estimate_study_hours({'difficulty': 'medium', 'topic': 'LangGraph I'})

[Observation] {"topic": "LangGraph I", "difficulty": "medium", "hours": 4}

[Final] Same content as ADK, again wrapped in the block-list format.
```

Prompt 3: I've completed weeks 1, 2, 3, 4, 6, and 7. What's next, what does the week after that cover, and how many hours should the next one take at easy difficulty?

ADK:

```
[Action] recommend_next_topics({'completed_weeks': [1, 2, 3, 4, 6, 7]})

[Observation] {'remaining_weeks': [8,9,10,12,13,14,15], 'next_week': 8, 'next_topic': 'Multi-agent systems'}

[Action] get_course_topic({'week': 9})

[Action] estimate_study_hours({'topic': 'Multi-agent systems', 'difficulty': 'easy'})

[Observation] {'week': 9, 'topic': 'LangGraph I'}
```

```
[Observation] {'topic': 'Multi-agent systems', 'difficulty': 'easy', 'hours': 2}
```

```
[Final] Next: Week 8 (Multi-agent systems). Then: Week 9 (LangGraph I). Easy-difficulty  
estimate for Week 8: ~2 hours.
```

LangGraph:

```
[Action] recommend_next_topics({'completed_weeks': [1, 2, 3, 4, 6, 7]})
```

```
[Observation] {"remaining_weeks": [8,9,10,12,13,14,15], "next_week": 8, "next_topic": "Multi-agent  
systems"}
```

```
[Action] get_course_topic({'week': 9})
```

```
[Action] estimate_study_hours({'topic': 'Multi-agent systems', 'difficulty': 'easy'})
```

```
[Observation] {"week": 9, "topic": "LangGraph I"}
```

```
[Observation] {"topic": "Multi-agent systems", "difficulty": "easy", "hours": 2}
```

```
[Final] Same content as ADK. Note: get_course_topic and estimate_study_hours were  
issued together in the same step by BOTH frameworks -- see NOTES.md.
```

NOTES.md — behavioral difference observed

Full write-up committed as week_6/NOTES.md. Summary: (1) ADK's final answer is a clean plain string; LangGraph/ChatGoogleGenerativeAI wraps it in a block-list structure with a 'signature' field, same wrapper difference already seen since Week 1/2 -- not agent-specific, just how that library represents any response. (2) Both frameworks parallelize independent tool calls in the same reasoning step rather than forcing one-at-a-time. (3) Both frameworks crash outright on an unhandled exception inside a tool body, instead of turning it into an error observation -- see Activity 2.

Activity 6 · Corte 2 Project: Pick Your Problem

For Corte 2, the plan is to extend Cubik-Lite into a multi-agent system (built with Google ADK or LangGraph) that keeps solving the same real problem — grounded calculus tutoring — but splits the work across specialized agents instead of one monolithic prompt. A triage agent would first classify the student's input (theory question vs. worked exercise, and which technique it likely involves) and route it accordingly; one or more solver/specialist agents would then handle the actual step-by-step resolution for their assigned technique (substitution, integration by parts, partial fractions, etc.), pulling grounding context from the existing RAG pipeline over ``data/``. This split lets each agent carry a narrower, technique-specific prompt instead of one system prompt trying to cover every case, and leaves room to add a verification role later. This will be refined once multi-agent patterns are covered in Week 8.

This description is submitting to the project to work on the future. This is the link of the Pull Request to merge: <https://github.com/JulianGuardian/Cubik-Lite/pull/5/changes>